

Examiner Marking Guidelines & Scoring Rubric

1. General Principles & Examination Rules

- **Strict Evaluation:** Award marks strictly based on scientific accuracy, clarity, and logical consistency.
- **Scoring Breakdown:**
 - **Full Marks:** Awarded for complete, scientifically accurate, and well-structured explanations.
 - **Partial Marks:** Awarded for incomplete or partially correct explanations where some accurate scientific concepts are demonstrated.
 - **Zero Marks (0):** Awarded for incorrect scientific knowledge, irrelevant answers, or complete omissions.
- **Step-by-Step Marking:** Examiners must mark each part step-by-step, evaluating the learner's response against specific rubrics rather than granting holistic impressions.
- **All the 2 items are compulsory, don't award for unanswered items.**
- **Be consistent throughout for all students strictly applying the same rules throughout, if you had marked incorrect for a particular answer, mark that incorrect throughout and don't change at any point.**
- **The RACE Competency Framework:** For every task/part, assess the following four core criteria:
 - **R – Relevance:** The candidate addresses the specific prompt, scenario context, and biology concepts directly.
 - **A – Accuracy:** Correct identification of scientific terms, processes, definitions, and causal relationships.
 - **C – Coherence:** Logical flow of thought, clear structure, and correct use of language and grammar.
 - **E – Excellence:** Demonstrates exceptional insight, detailed biological mechanics, and meets all criteria thoroughly.

2. Detailed Marking Breakdown

Item 1: Nutrition, Kwashiorkor, and Digestion (Total: 20 Marks)

- **Part (a) [8 Marks]:**
 - **Identification (2 Marks):** Correctly identifies Kwashiorkor.
 - **Explanation of Damage (6 Marks):** Explains the biological impact of a pure starch diet lacking protein (e.g., fluid retention/edema leading to swollen belly, muscle wasting, hair pigmentation changes, lack of amino acids for tissue repair and growth).
 - *RACE Application:* Full 8 marks require accurate identification + thorough biological explanation + well-structured presentation.
- **Part (b) [12 Marks]:**
 - **Importance of Protein in Children (6 Marks):** Detailed explanation of protein functions in growing children (growth, tissue repair, enzyme/hormone synthesis, immune antibody production).
 - **Digestion of Proteins (6 Marks):** Step-by-step pathway showing how a child's stomach and small intestine break down proteins into amino acids (pepsin, trypsin, peptidases, pH environments), refuting the mothers' misconception.

Item 2: Water Movement in Plants & Osmosis (Total: 20 Marks)

- **Part (a) [8 Marks]:**
 - **Osmotic Mechanism (8 Marks):** Full explanation of how excess dry fertilizer and salty water lower the solute potential (Ψ_s) / water potential (Ψ_w) of soil water relative to the cell sap of root hair cells. Explains hypertonic soil environment, movement of water out of roots via osmosis (exosmosis), loss of turgor pressure, and subsequent wilting.
- **Part (b) [12 Marks]:**
 - **Effect of Leaf Spraying (6 Marks):** Explanation of chemical damage, stomatal blockage/damage, acceleration of desiccation, or localized hypertonic stress worsening leaf damage.
 - **Correct Farming Advice (6 Marks):** Actionable, scientifically sound guidelines for watering and fertilization (e.g., diluting fertilizers, using fresh non-salty water, applying fertilizer at correct distances/amounts, proper soil testing/irrigation).

Detailed Model Answers & Exemplars**ITEM 1 MODEL ANSWERS****Part (a): Identification and Pathophysiology of Starch-Only Diet (8 Marks)****Correct Answer (Full Marks Exemplar)**

Condition Name: Kwashiorkor.

Explanation: Kwashiorkor is a severe form of protein-energy malnutrition that occurs when a child's diet consists almost entirely of carbohydrates (such as cassava porridge and sweet potatoes) without sufficient protein intake.

When a child eats only starchy foods:

1. **Fluid Imbalance (Edema):** Protein is essential for synthesizing blood plasma proteins like albumin. A lack of protein reduces blood osmotic pressure, causing plasma fluid to leak into surrounding interstitial tissues. This leads to water retention, causing the characteristic **swollen belly** (ascites) and edema in limbs.
2. **Impaired Growth & Muscle Wasting:** Proteins provide amino acids necessary for building cellular structures, tissue repair, and muscle growth. Without amino acids, body proteins are broken down to sustain vital organs, leading to **weak bodies** and stunted growth.
3. **Pigmentation & Skin Damage:** Melanin (the pigment responsible for skin and hair color) requires amino acids like tyrosine for synthesis. A deficiency leads to **reddish/lightened hair** and **pale, flaky skin**.
4. **Anemia:** Proteins are vital for hemoglobin formation. Insufficient protein impairs red blood cell production, resulting in **pale skin** due to anemia.

Incorrect Answer (Zero/Low Marks Exemplar)

The children are suffering from Marasmus or bewitchment. The children are weak because cassava and sweet potatoes are bad foods that poison the stomach. When a child stops breastfeeding, their stomach cannot handle cassava, so it boils inside and turns the hair red. They should drink traditional herbs to cure the belly.

Examiner Note: Incorrect diagnosis (Marasmus/witchcraft), incorrect biology, non-scientific rationale. Zero marks.

Part (b): Functions of Protein & Protein Digestion Pathway (12 Marks)**Correct Answer (Full Marks Exemplar)****Why Growing Children Need Protein Foods (6 Marks):**

1. **Growth and Structural Development:** Children undergo rapid cell division. Proteins supply amino acids required to synthesize cellular membranes, structural tissues, muscles, and bones.
2. **Enzyme and Hormone Synthesis:** Biological catalysts (enzymes) and chemical messengers (such as growth hormone and insulin) are proteins needed for metabolic processes.

3. **Immune System Function:** Antibodies that fight infections are proteins. Protein-deficient children are susceptible to frequent illnesses.
4. **Tissue Repair:** Daily activity leads to tissue wear and tear; proteins are necessary to repair damaged body tissues.

How a Child's Digestive System Breaks Down Proteins (6 Marks): Contrary to the belief that young children cannot digest thick protein foods like groundnut paste, a child's digestive system produces specific enzymes tailored to break proteins down into simple, absorbable amino acids:

1. **Stomach:**
 - Mechanical digestion begins with chewing, and swallowing delivers the protein to the stomach.
 - Gastric juice secreted in the stomach contains **hydrochloric acid (HCl)** and inactive pepsinogen.
 - HCl creates an acidic pH (~1.5–2.0), activating pepsinogen into **pepsin**.
 - Pepsin breaks long protein chains down into smaller polypeptides.
2. **Small Intestine (Duodenum & Ileum):**
 - Pancreatic juice releases **trypsin** into the duodenum (in an alkaline pH maintained by bile/bicarbonate). Trypsin digests polypeptides into shorter peptides.
 - Epithelial glands in the ileum secrete **peptidases** (erepsin), which complete digestion by breaking peptides down into single **amino acids**.
3. **Absorption:**
 - The resulting amino acids are small, soluble molecules absorbed across the villi of the ileum into the bloodstream via active transport/diffusion and delivered to cells throughout the body. Thus, children are fully equipped to digest foods like groundnut paste.

Incorrect Answer (Zero/Low Marks Exemplar)

Children need protein so they don't die. The stomach cannot break down groundnut paste because groundnut paste is too thick and clogs the intestine. The stomach juice melts cassava but groundnuts just sit in the stomach and cause pain.

Examiner Note: Fails to describe enzyme actions (pepsin, trypsin, peptidases), provides incorrect digestive mechanics, and validates false cultural myths without scientific basis.

ITEM 2 MODEL ANSWERS

Part (a): Mechanism of Exosmosis and Wilting (8 Marks)

Correct Answer (Full Marks Exemplar)

Osmotic Explanation of Wilting:

1. **Creation of a Hypertonic Solution:** Applying a large quantity of dry artificial fertilizer combined with salty water significantly increases the concentration of solute ions (salts and nitrates) in the soil solution.
2. **Water Potential Gradient:** The soil water becomes **hypertonic** relative to the cell sap inside the root hair cells. This means the soil solution has a lower water potential (\$Ψ\$) / higher solute concentration than the internal environment of the plant's root hair cells.
3. **Exosmosis:** Water moves along a water potential gradient from a region of higher water potential (inside the root hair cells) to a region of lower water potential (the hypertonic soil) across the semi-permeable cell membrane via **osmosis (exosmosis)**.
4. **Loss of Turgidity:** As water rapidly leaves the root cells, the vacuole shrinks, and the plasma membrane pulls away from the cell wall (plasmolysis).

5. **Wilting:** Root cells lose **turgor pressure**, which normally provides mechanical support to non-woody stem tissues. As a result, the stems and leaves lose firmness and droop/wilt, despite the soil being physically wet.

Incorrect Answer (Zero/Low Marks Exemplar)

The fertilizer burned the roots because it was too hot. Salty water blocks the holes in the roots so water cannot go inside the plant. The water stayed in the soil because the fertilizer sucked it up like a sponge.

Examiner Note: Uses non-scientific concepts ("burned," "blocked holes," "sucked up like a sponge") without referencing osmosis, water potential, or hypertonic solutions.

Part (b): Chemical Damage and Farmer Advice (12 Marks)**Correct Answer (Full Marks Exemplar)****Why Spraying Leaves Made Damage Worse (6 Marks):**

1. **Misdiagnosis & Additional Stress:** The plant was suffering from osmotic desiccation (water loss), not pest attack.
2. **Stomatal Blockage & Direct Leaf Toxicity:** Spraying high concentrations of strong chemicals directly on wilted leaves damages the fragile epidermal and mesophyll tissues.
3. **Chemical Hypertonicity:** The chemical spray deposited additional solutes on leaf surfaces, drawing remaining moisture out of leaf epidermal cells via osmosis.
4. **Disruption of Transpiration & Gas Exchange:** Toxic chemical build-up can force stomata shut or destroy guard cells, preventing photosynthesis and gas exchange while accelerating tissue necrosis (drying up and death).

Correct Watering and Fertilization Advice for the Future (6 Marks):

1. **Use Fresh, Non-Salty Water:** Water crops using fresh river or rainwater rather than saline deep-well water to prevent salt accumulation in the soil.
2. **Proper Fertilizer Application Rate:** Apply artificial fertilizers in strictly controlled, recommended quantities; avoid excessive application.
3. **Placement:** Apply dry fertilizer a few centimeters away from the plant stem (e.g., in a ring around the drip line) rather than directly against the stem, preventing localized high salt concentrations.
4. **Dissolve/Irrigate Appropriately:** Apply fertilizer when soil is already moist or thoroughly irrigate with clean water immediately after application to dilute solute concentration and facilitate balanced absorption.
5. **Accurate Problem Diagnosis:** Check soil moisture and root osmotic conditions before spraying pesticides unnecessarily.

Incorrect Answer (Zero/Low Marks Exemplar)

The spray killed the leaves because the farmer sprayed during the day instead of at night. Next time, the farmer should use more salty water to wash away the chemical spray. He should put fertilizer directly on top of the leaves next time.

Examiner Note: Contains scientifically detrimental advice, fails to explain leaf desiccation, and misunderstands proper agricultural methods. Zero marks.